Spectral RMS Is Not Generally Equal to Averaged Curvature

An Exact Spectral–Phase Identity and a Closed-Curve Counterexample

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Abstract

For a smooth, nonvanishing eigenmode of the periodic operator $-\partial_s^2$, we establish an exact decomposition of its eigenvalue into squared weighted mean phase gradient, weighted phase variance, and normalized amplitude-gradient energy. The decomposition yields sharp equality conditions and does not require a nonnegative phase gradient. An explicit eigenmode with $\sqrt{\lambda} = 1$ and weighted mean phase gradient $4/5$ can moreover be realized as the unit tangent of a smooth closed planar curve, where the phase gradient equals its curvature. The conclusion applies to the declared free periodic operator, without assuming a derived physical confinement or mass operator.

1 Setting and exact identity

Let $S^1_L = \mathbb{R}/L\mathbb{Z}$ denote a circle of length $L > 0$. Consider the self-adjoint free operator

$$H = -\frac{\mathrm{d}^2}{\mathrm{d}s^2}, \qquad D(H) = H^2_{\mathrm{per}}(S^1_L;\mathbb{C}). \tag{1}$$

Suppose that χ is a smooth, nowhere-vanishing periodic eigenfunction:

$$-\chi''(s) = \lambda\chi(s), \qquad \chi(s) \neq 0. \tag{2}$$

Because χ has no zeros, its polar form may be written on the universal cover as

$$\chi(s) = R(s)e^{i\phi(s)}, \qquad R(s) > 0, \qquad \sigma(s) = \phi'(s) \in \mathbb{R}. \tag{3}$$

Although ϕ need not be periodic, ϕ' is periodic, and $\phi(s+L) - \phi(s) = 2\pi n$ for some $n \in \mathbb{Z}$. Set

$$N = \int_0^L R(s)^2\,\mathrm{d}s > 0, \qquad w(s) = \frac{R(s)^2}{N}, \qquad \langle f\rangle_w = \int_0^L f(s)w(s)\,\mathrm{d}s, \tag{4}$$

and define

$$\mathrm{Var}_w(\sigma) = \langle\sigma^2\rangle_w - \langle\sigma\rangle_w^2 = \int_0^L w(s)\big(\sigma(s) - \langle\sigma\rangle_w\big)^2\,\mathrm{d}s. \tag{5}$$

Theorem 1 (Sharp spectral–phase decomposition)**.** *Under the hypotheses above,*

$$\boxed{\lambda - \langle\sigma\rangle_w^2 = \frac{\displaystyle\int_0^L |R'(s)|^2\,\mathrm{d}s}{\displaystyle\int_0^L R(s)^2\,\mathrm{d}s} + \mathrm{Var}_w(\sigma) \geq 0.} \tag{6}$$

Equality in (6) holds if and only if both R and σ are constant. In particular,

$$\sqrt{\lambda} \geq |\langle \sigma \rangle_w|. \tag{7}$$

No sign restriction on σ is needed.

Proof. Multiply (2) by $\overline{\chi}$ and integrate:

$$\lambda \int_0^L |\chi|^2 \, \mathrm{d}s = -\int_0^L \overline{\chi}\, \chi'' \, \mathrm{d}s. \tag{8}$$

Periodic integration by parts, using periodicity of both χ and χ', eliminates the boundary term and yields

$$\lambda N = \int_0^L |\chi'|^2 \, \mathrm{d}s. \tag{9}$$

From (3),

$$\chi' = e^{i\phi}(R' + iR\sigma), \qquad |\chi'|^2 = |R'|^2 + R^2\sigma^2, \tag{10}$$

since R' and σ are real. Inserting this expression into (9) and dividing by N gives

$$\lambda = \frac{\int_0^L |R'|^2 \, \mathrm{d}s}{N} + \langle \sigma^2 \rangle_w. \tag{11}$$

By the weighted variance identity, $\langle \sigma^2 \rangle_w = \langle \sigma \rangle_w^2 + \mathrm{Var}_w(\sigma)$. Substitution into (11) proves (6). Both terms on its right-hand side are nonnegative, proving (7).

For the equality case, the sum in (6) vanishes if and only if both of its nonnegative terms vanish. As R is smooth, $\int_0^L |R'|^2 \, \mathrm{d}s = 0$ implies $R' = 0$ everywhere, hence R is constant. Likewise, $w > 0$ everywhere and smoothness of σ show that $\mathrm{Var}_w(\sigma) = 0$ implies $\sigma(s) = \langle \sigma \rangle_w$ everywhere. The converse follows by direct substitution, completing the proof. □

Remark 1 (Winding and the sign of the mean)**.** *When equality holds, $\chi(s) = R_0 e^{i(\sigma_0 s + \phi_0)}$, where $\sigma_0 L = 2\pi n$, $n \in \mathbb{Z}$, and $\lambda = \sigma_0^2$. Consequently, equality in $\lambda = \langle \sigma \rangle_w^2$ allows either sign of the constant phase gradient. The stronger unsigned statement $\sqrt{\lambda} = \langle \sigma \rangle_w$ requires in addition $\langle \sigma \rangle_w \geq 0$; without this assumption the exact equality is $\sqrt{\lambda} = |\langle \sigma \rangle_w|$.*

2 An explicit strict counterexample

Proposition 1 (Nonconstant amplitude defeats mean closure)**.** *For $s \in [0, 2\pi]$, let*

$$\chi_a(s) = \cos s + ia \sin s, \qquad a > 0. \tag{12}$$

Then χ_a is a smooth, nowhere-vanishing periodic eigenfunction with $\lambda = 1$, while its weighted mean phase gradient is

$$\langle \sigma \rangle_w = \frac{2a}{1 + a^2}. \tag{13}$$

Thus $\sqrt{\lambda} \neq \langle \sigma \rangle_w$ for every $a > 0$ with $a \neq 1$. In particular, at $a = \frac{1}{2}$,

$$\boxed{\sqrt{\lambda} = 1, \qquad \langle \sigma \rangle_w = \frac{4}{5}.} \tag{14}$$

Proof. Direct differentiation yields $-\chi_a'' = \chi_a$, hence $\lambda = 1$. Moreover,

$$R_a^2(s) = |\chi_a(s)|^2 = \cos^2 s + a^2 \sin^2 s \geq \min\{1, a^2\} > 0. \tag{15}$$

The normalization is

$$N_a = \int_0^{2\pi} R_a^2(s)\, \mathrm{d}s = \pi(1 + a^2). \tag{16}$$

For a nonvanishing complex function, its phase derivative is

$$\sigma_a(s) = \frac{\mathrm{Im}(\overline{\chi_a(s)}\chi_a'(s))}{|\chi_a(s)|^2}. \tag{17}$$

Because $\mathrm{Im}(\overline{\chi_a}\chi_a') = a$, we have

$$\sigma_a(s) = \frac{a}{R_a^2(s)} > 0. \tag{18}$$

Therefore $R_a^2\sigma_a = a$ pointwise and

$$\langle\sigma\rangle_w = \frac{1}{N_a}\int_0^{2\pi} R_a^2\sigma_a\, \mathrm{d}s = \frac{2\pi a}{\pi(1+a^2)} = \frac{2a}{1+a^2}. \tag{19}$$

The difference

$$1 - \frac{2a}{1+a^2} = \frac{(a-1)^2}{1+a^2} \tag{20}$$

is strictly positive unless $a = 1$. Taking $a = 1/2$ gives (14). □

Exact gap accounting at $a = 1/2$. For the explicit example $R^2 = \cos^2 s + \frac{1}{4}\sin^2 s$ and $N = 5\pi/4$. The elementary integral

$$\int_0^{2\pi} \frac{\mathrm{d}s}{\cos^2 s + \frac{1}{4}\sin^2 s} = 4\pi \tag{21}$$

gives, using $\sigma = (2R^2)^{-1}$,

$$\langle\sigma^2\rangle_w = \frac{1}{4N}\int_0^{2\pi} \frac{\mathrm{d}s}{R^2(s)} = \frac{4}{5}. \tag{22}$$

Hence

$$\mathrm{Var}_w(\sigma) = \frac{4}{5} - \frac{16}{25} = \frac{4}{25}, \tag{23}$$

while (11) yields

$$\frac{\int_0^{2\pi} |R'|^2\, \mathrm{d}s}{N} = 1 - \frac{4}{5} = \frac{1}{5}. \tag{24}$$

Thus both contributions to the gap are explicit:

$$\boxed{1 - \left(\frac{4}{5}\right)^2 = \underbrace{\frac{1}{5}}_{\text{amplitude-gradient energy}} + \underbrace{\frac{4}{25}}_{\text{phase variance}} = \frac{9}{25}.} \tag{25}$$

3 Realization by a smooth closed planar curve

Proposition 2 (Geometrically locked counterexample)**.** *For $a > 0$, define the complex unit tangent*

$$T_a(s) = \frac{\chi_a(s)}{|\chi_a(s)|} \tag{26}$$

and the plane curve, with $\mathbb{C} \cong \mathbb{R}^2$,

$$\Gamma_a(s) = \int_0^s T_a(t)\,\mathrm{d}t. \tag{27}$$

Then Γ_a is a smooth, closed, unit-speed planar curve of length 2π. Its curvature κ_a and torsion τ_a satisfy

$$\kappa_a(s) = \sigma_a(s) > 0, \qquad \tau_a(s) = 0, \tag{28}$$

and hence the pointwise geometric lock

$$\sigma_a(s) = \sqrt{\kappa_a(s)^2 + \tau_a(s)^2} \tag{29}$$

holds. Nevertheless, at $a = 1/2$ the associated free periodic eigenmode has $\sqrt{\lambda} = 1$ but $\langle\kappa\rangle_w = 4/5$.

Proof. Smoothness follows from $|\chi_a| > 0$, and $|\Gamma_a'| = |T_a| = 1$. Since $\chi_a(s+\pi) = -\chi_a(s)$ and $R_a(s+\pi) = R_a(s)$, we have $T_a(s+\pi) = -T_a(s)$. It follows that

$$\Gamma_a(2\pi) - \Gamma_a(0) = \int_0^\pi \left(T_a(s) + T_a(s+\pi)\right)\mathrm{d}s = 0. \tag{30}$$

The curve is closed. Writing $T_a = e^{i\phi_a}$,

$$T_a' = i\phi_a' T_a = i\sigma_a T_a, \tag{31}$$

so its curvature is $\kappa_a = |T_a'| = |\sigma_a| = \sigma_a$, where positivity follows from (18). As the curve is planar, its torsion is zero. The final assertion follows from Proposition 1. □

4 Scope and consequences

The exact relation can be written in spectral-root form as

$$\boxed{\sqrt{\lambda} = \left(\langle\sigma\rangle_w^2 + \mathrm{Var}_w(\sigma) + \frac{\|R'\|_{L^2}^2}{\|R\|_{L^2}^2}\right)^{1/2}.} \tag{32}$$

This demonstrates that the spectral scale $\sqrt{\lambda}$ cannot generally be replaced by the weighted mean $\langle\sigma\rangle_w$, even if a pointwise identification of σ with geometric curvature has been independently supplied.

If the separated-field dispersion of an independently specified product-space model yields a conditional effective mass $m_{\mathrm{eff}} = (\hbar/c)\sqrt{\lambda}$, then replacing this with $(\hbar/c)\langle\sigma\rangle_w$ demands the equality hypotheses, together with the appropriate sign convention. This theorem does not derive the confinement operator from an underlying physical action, identify particle modes, or establish an empirical mass spectrum. Operators with nonzero potentials, connections, or different boundary conditions may require a different energy identity; zero-containing eigenmodes require additional treatment at their nodes.

Source and context. See the project note `GEOMETRIC_MASS_CLOSURE.md`.